

Hypothesis tests, effects, practical significance, A/B guardrails

BEFORE-CLASS MIND MAP

Quantitative Methods for Business Analysis

About 10 minutes · read before class

HOW TO USE IT

1

Start with the big picture on page 2: how the 3 sections fit together.

2

Open each section map. Read each idea and its short note, and follow the labelled lines.

3

Answer the before-class question at the foot of each map and bring your answers to class.

INSIDE

S06

The big picture

How do we decide if a business change truly works?

p. 2

6.1

Hypothesis Testing Logic, Errors, and Benchmark Mean Comparisons

How do we know if a sample mean beats a goal?

p. 3

6.2

A/B Testing with Proportions and Statistical Uplift

How do we test whether a new version improves conversion?

p. 4

6.3

Practical Significance, Guardrail Metrics, and Cross-Tool Reconciliation

When is a statistically detectable improvement safe to launch?

p. 5

READING THE MAP



Idea



Idea written as a formula



Watch for this in class

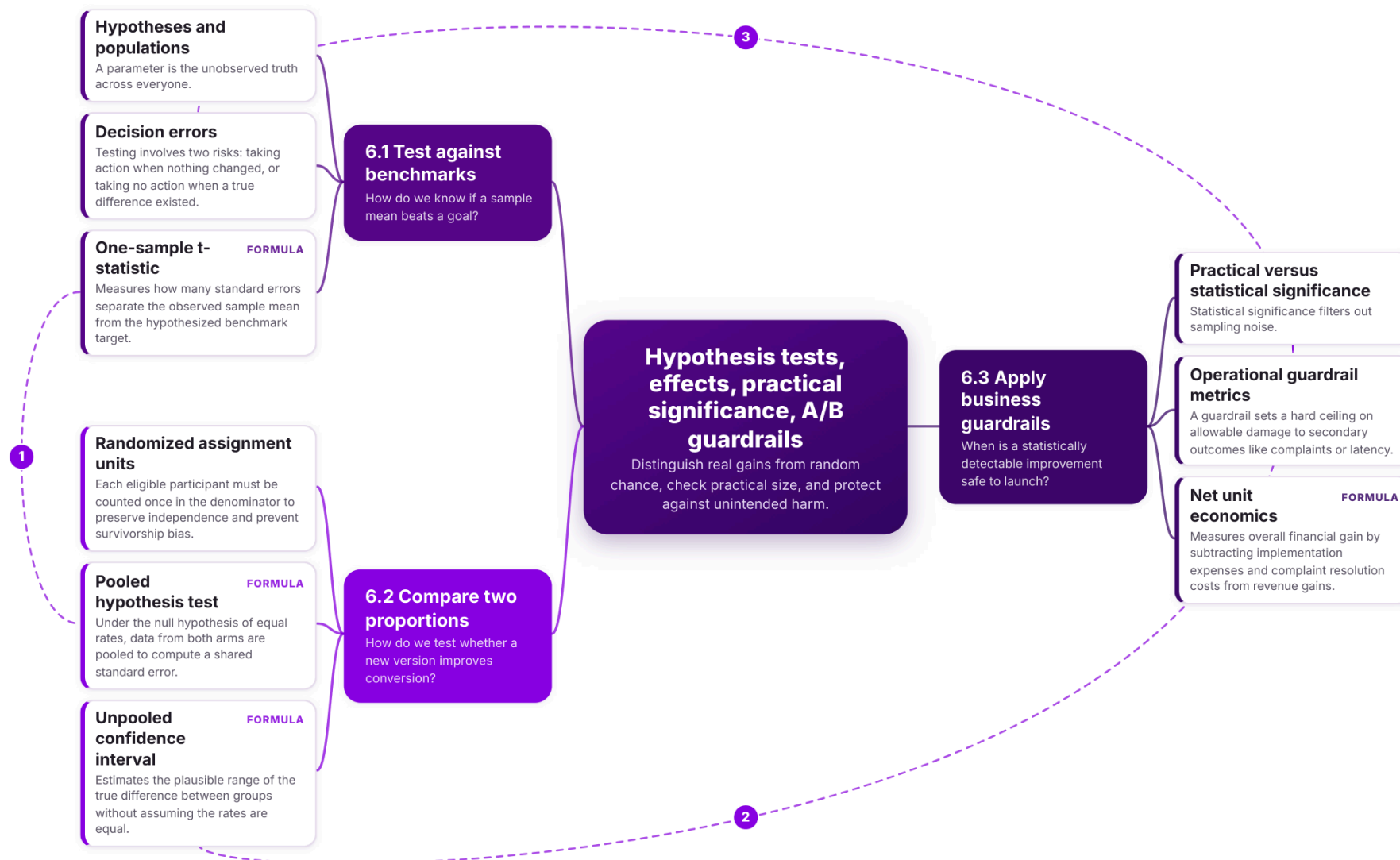


Everyday example

Bring from earlier sessions: Conditional risk, expected value, simulation (Session 4).

Learners test performance claims against benchmarks, evaluate randomized experiments, and apply guardrail metrics to rollout decisions.

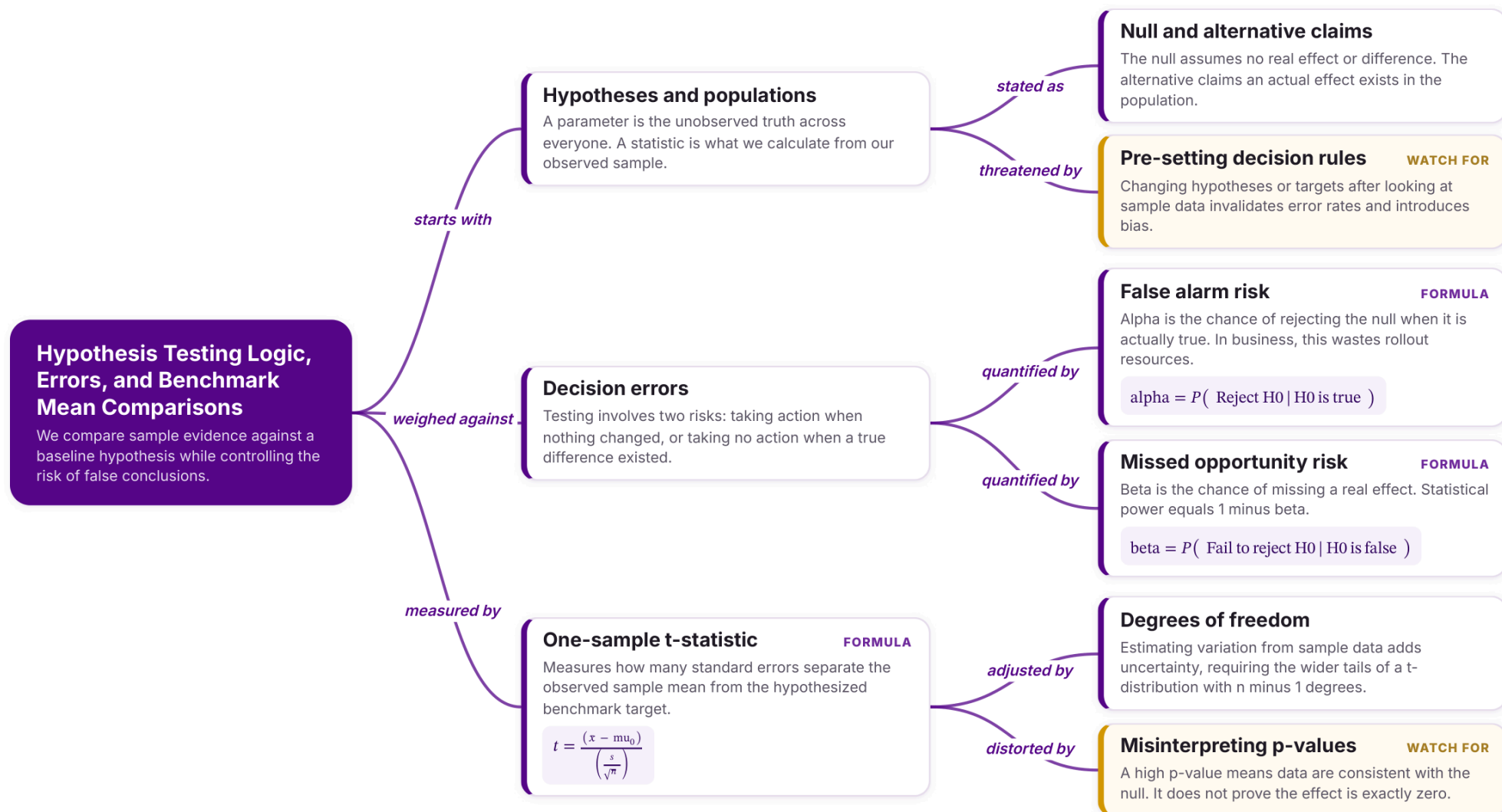
Question: How do we decide if a business change truly works?

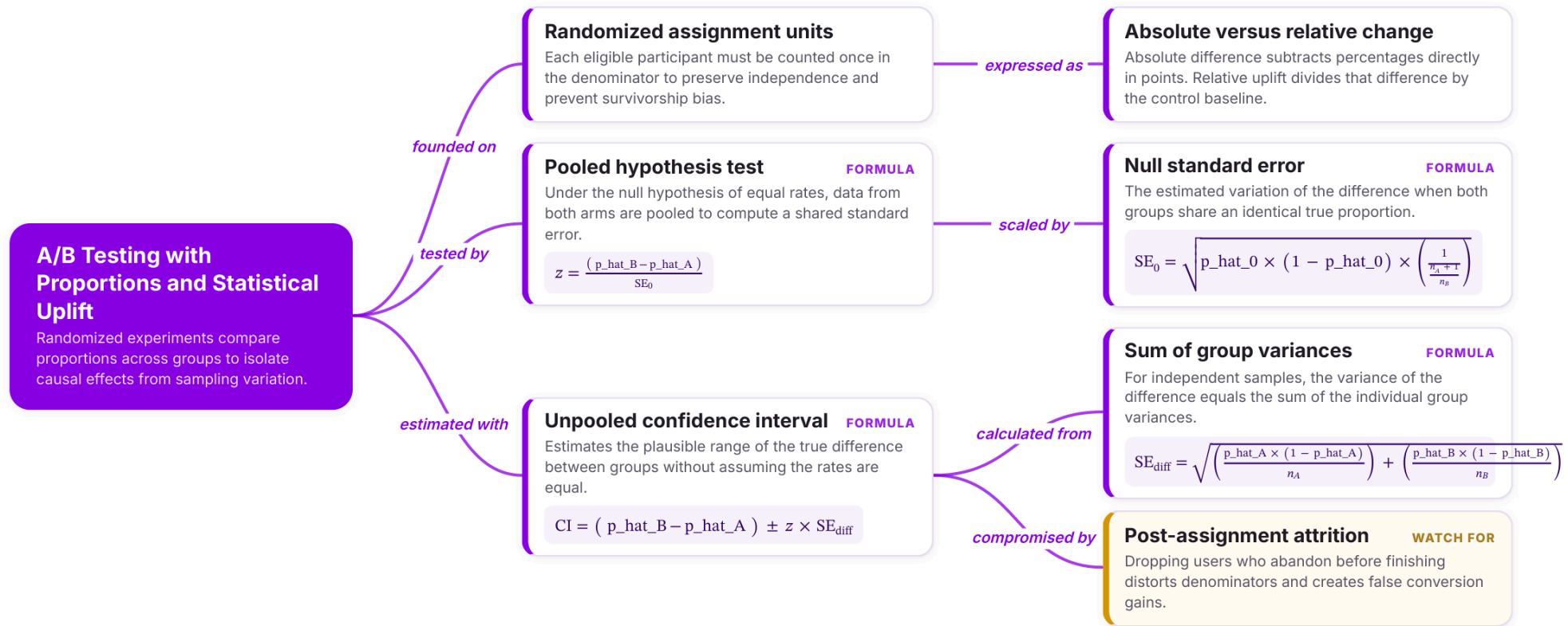


1 One-sample t-statistic → Pooled hypothesis test: standardizing differences against noise

2 Unpooled confidence interval → Practical versus statistical significance: plausible ranges meet business hurdles

3 Decision errors → Operational guardrail metrics: decision rules balance operational risks





Practical Significance, Guardrail Metrics, and Cross-Tool Reconciliation

Question: When is a statistically detectable improvement safe to launch?

In class: **slides 53–73** · Lab Practice 3
After class: Toolkit **pp. 45–68** · Concise pp. 16–22

